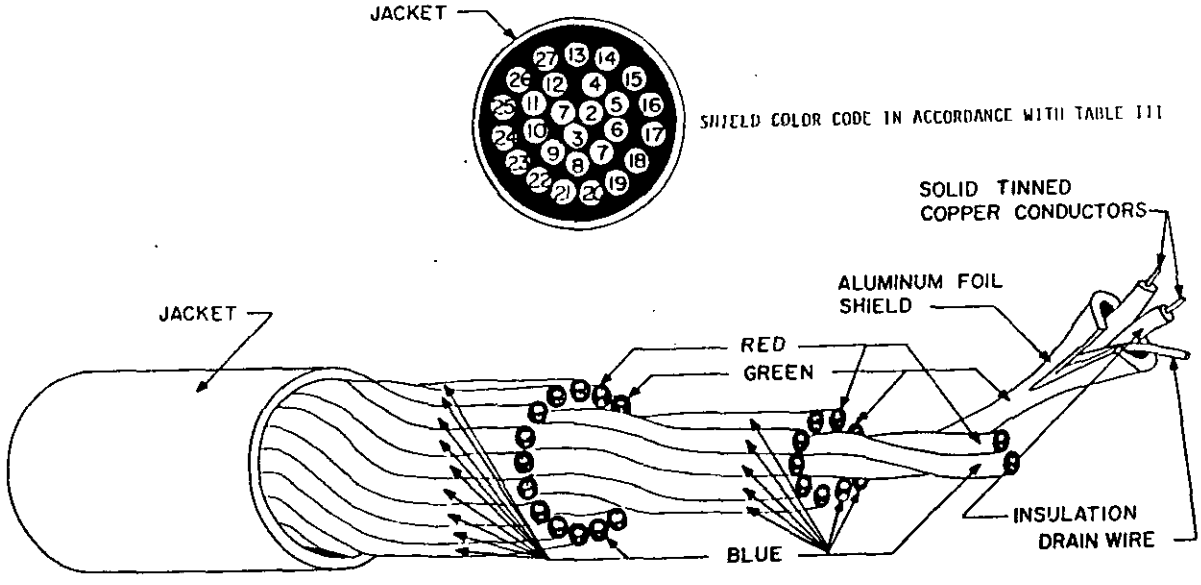
MILITARY SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, TWENTY-SEVEN INDIVIDUALLY SHIELDED PAIRS OF 24 (7 X 32) AWG

NOTE: NOT FOR AEROSPACE USE.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.




27 PAIRS INDIVIDUALLY SHIELDED

FIGURE 1. Cable illustration.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Engineering information: (See table I).

TABLE I. Description, electrical.

| Electrical characteristics | Requirements |
|---|---|
| Nominal capacitance between conductors | 12 pF/ft (picofarads/foot) |
| Nominal capacitance between the conductor and the shield with one conductor tied to the shield | 23 pF/ft |
| Continuous working voltage | 300 V rms maximum between conductors |
| Nominal capacitance between adjacent shields | 150 pF/ft |
| Continuous working voltage between adjacent shields | 50 volts maximum |
| Shield dc resistance | 20.5 ohms/1,000 feet maximum |

TABLE II. Description, physical.

| Components | Construction details |
|---|---|
| Number of pairs | Twenty-seven individually shielded |
| Conductor type and wire size | Stranded tinned copper 24 AWG |
| Conductor stranding | 7 X 32 AWG |
| Conductor insulation | Expanded polyethylene |
| Conductor insulation thickness | 0.023 inch nominal thickness |
| Drain wire type and size | Stranded tinned copper 24 AWG |
| Drain wire stranding | 16 X 36 AWG |
| Jacket material | Polyvinyl chloride |
| Jacket thickness | 0.087 inch |
| Finished cable diameter | 0.919 inch maximum outside diameter |
| Cable style (UL) | 2493 |
| Tensile strength (jacket) | 2,000 pounds per square inch minimum |
| Elongation (jacket) | 150 percent minimum |
| Overall cabling lay lengths | 1.3 twists per foot ±10 percent |



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The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).

TABLE III. Shield color code.

| Pair number | Shield color |
|---|---|
| *Center core* | |
| 1st pair | Red |
| 2nd pair | Green |
| 3rd pair | Blue - 1 |
| Inner layer | |
| 4th pair | Red |
| 5th pair | Green |
| 6th pair | Blue - 1 |
| 7th pair | Blue - 2 |
| 8th pair | Blue - 3 |
| 9th pair | Blue - 4 |
| 10th pair | Blue - 5 |
| 11th pair | Blue - 6 |
| 12th pair | Blue - 7 |
| Outer layer | |
| 13th pair | Red |
| 14th pair | Green |
| 15th pair | Blue - 1 |
| 16th pair | Blue - 2 |
| 17th pair | Blue - 3 |
| 18th pair | Blue - 4 |
| 19th pair | Blue - 5 |
| 20th pair | Blue - 6 |
| 21th pair | Blue - 7 |
| 22th pair | Blue - 8 |
| 23th pair | Blue - 9 |
| 24th pair | Blue - 10 |
| 25th pair | Blue - 11 |
| 26th pair | Blue - 12 |
| 27th pair | Blue - 13 |

REQUIREMENTS:

*Design and construction:* (See tables II and III).

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact.

Drain wire location. Each drain wire is to be spirally located beneath the shield with the insulated conductors. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.



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Flammability. The cable shall pass UL 1581 VW-1 flame test requirements.

Shield integrity test.*

| Frequency | Response ** |
|---|---|
| 100 kHz | 110 dB below reference |
| 500 kHz | 105 dB below reference |
| 1 MHz | 100 dB below reference |
| 5 MHz | 86 dB below reference |
| 10 MHz | 79 dB below reference |
| 15 MHz | 74 dB below reference |
| 20 MHz | 69 dB below reference |
| 25 MHz | 65 dB below reference |
| 30 MHz | 61 dB below reference |

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 42 dB below reference.

Crosstalk test limits. All values are minimums.

| Frequency (KHz) | Near end crosstalk (dB) | Far end crosstalk (dB) |
|---|---|---|
| 40 | 63 | 63 |
| 70 | 59 | 58 |
| 100 | 56 | 56 |
| 1,000 | 56 | 54 |

Insulation resistance between shields: 100 megohms/1,000 feet (minimum).

Durometer hardness. The cable jacket shall have a "Shore A" hardness of 82 ±5.

Part or Identifying Number (PIN): The PIN shall be M49285/33.

## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Air Force - 71
DLA - ES, IS

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1114-34)



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